Name : Nisha Yuvraj Pragane | DIV:SEDA | Roll no :13 | PRN : 1252090013 | BATCH :B1 A retail company wants to group customers based on Annual Income and Spending Score for targeted marketing. Dataset: Mall Customer Segmentation (Kaggle): https://www.kaggle.com/datasets/shwetabh123/mall-customers a. Preprocess data (missing values, encoding, scaling) b. Apply Hierarchical Clustering and plot dendrography c. Visualize clusters (Income vs Spending Score d. Interpret and describe each customer segment

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.cluster import KMeans
from sklearn.preprocessing import StandardScaler
from sklearn.metrics import silhouette_score
from sklearn.model_selection import train_test_split

from scipy.cluster.hierarchy import dendrogram, linkage
from sklearn.cluster import AgglomerativeClustering
```

Load Dataset

```
df = pd.read_csv('Iris.csv')
df.head()
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa

Next steps: Generate code with df New interactive sheet

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 6 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Id             150 non-null    int64
 1   SepalLengthCm  150 non-null    float64
 2   SepalWidthCm   150 non-null    float64
 3   PetalLengthCm  150 non-null    float64
 4   PetalWidthCm   150 non-null    float64
 5   Species        150 non-null    object
dtypes: float64(4), int64(1), object(1)
memory usage: 7.2+ KB
```

```
df.describe()
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
count	150.000000	150.000000	150.000000	150.000000	150.000000
mean	75.500000	5.843333	3.054000	3.758667	1.198667
std	43.445368	0.828066	0.433594	1.764420	0.763161
min	1.000000	4.300000	2.000000	1.000000	0.100000
25%	38.250000	5.100000	2.800000	1.600000	0.300000
50%	75.500000	5.800000	3.000000	4.350000	1.300000
75%	112.750000	6.400000	3.300000	5.100000	1.800000
max	150.000000	7.900000	4.400000	6.900000	2.500000

Data Preprocessing

```
df = df.dropna()
```

Feature Scaling

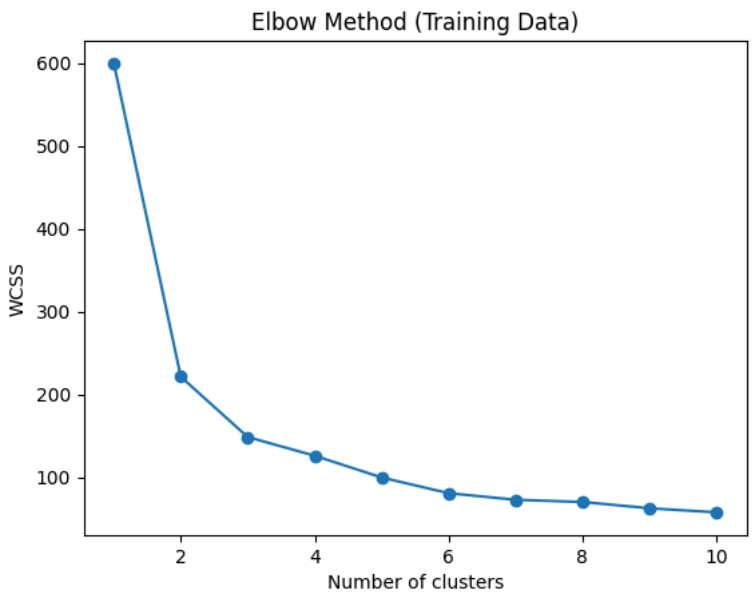
```
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
X_train, X_test = train_test_split(X_scaled, test_size=0.2, random_state=42)
```

Elbow Method

```
wcss = []

for i in range(1, 11):
    kmeans = KMeans(n_clusters=i, random_state=42)
    kmeans.fit(X_train)
    wcss.append(kmeans.inertia_)

plt.plot(range(1, 11), wcss, marker='o')
plt.title("Elbow Method (Training Data)")
plt.xlabel("Number of clusters")
plt.ylabel("WCSS")
plt.show()
```



K-Means Clustering

```
kmeans = KMeans(n_clusters=2, random_state=42)
kmeans.fit(X_train)
y_train = kmeans.predict(X_train)
y_test = kmeans.predict(X_test)
```

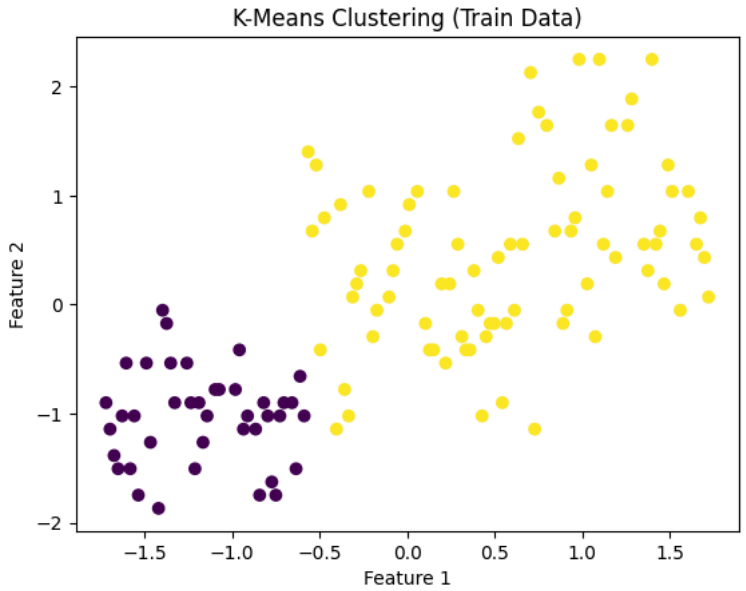
Silhoueete Score

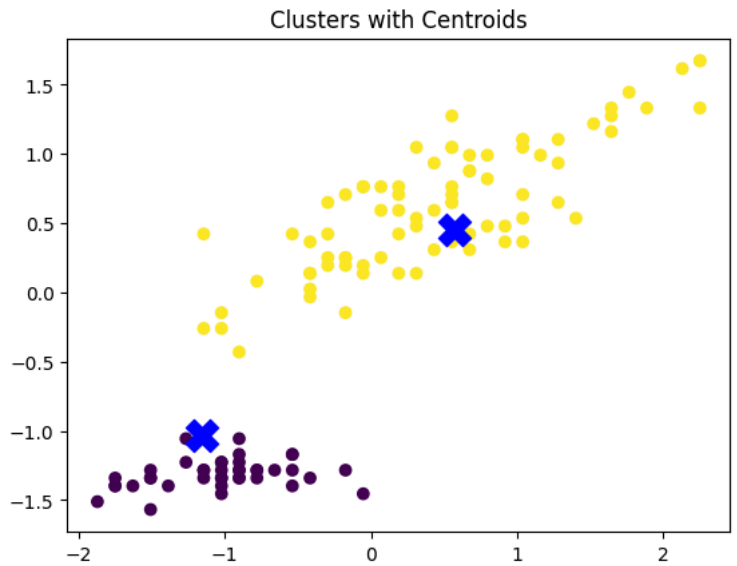
```
train_score = silhouette_score(X_train, y_train)
test_score = silhouette_score(X_test, y_test)
print("Train Silhouette Score:", train_score)
print("Test Silhouette Score:", test_score)
```

```
Train Silhouette Score: 0.5678057687378739
Test Silhouette Score: 0.5755796052349909
```

Visualization

```
plt.scatter(X_train[:, 0], X_train[:, 1], c=y_train, cmap='viridis')
plt.title("K-Means Clustering (Train Data)")
plt.xlabel("Feature 1")
plt.ylabel("Feature 2")
plt.show()
```




```
centroids = kmeans.cluster_centers_

plt.scatter(X_train[:, 0], X_train[:, 1], c=y_train, cmap='viridis')
plt.scatter(centroids[:, 0], centroids[:, 1], s=300, c='blue', marker='X')
plt.title("Clusters with Centroids")
plt.show()
```